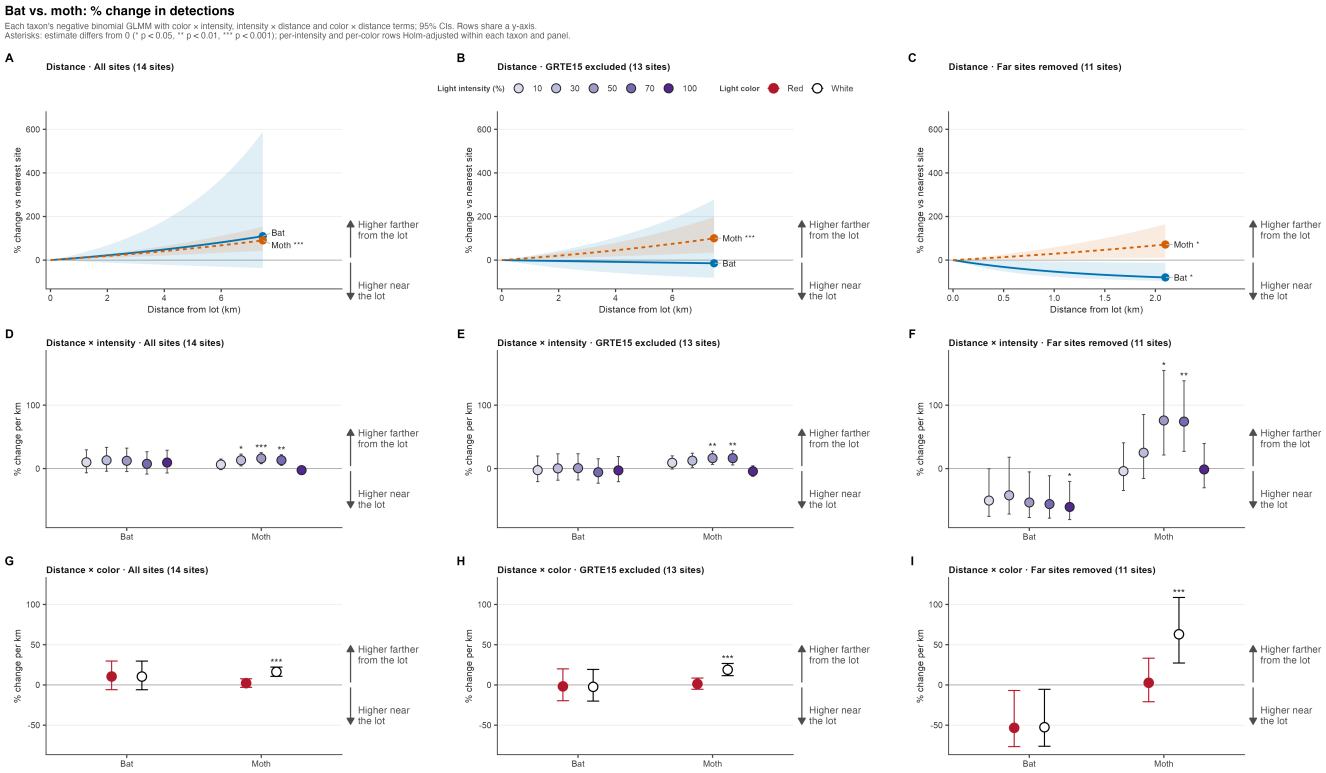


Figure 1: Bat (blue circles, solid lines) vs. moth (orange circles, dashed lines) % change in detections; color-blind-safe Okabe–Ito colors. Columns, left to right: all 14 sites; all sites except GRTE15 (13 sites); far sites (GRTE13–15, ~7 km) removed (11 sites, 0–2.1 km). **A**–**C** distance: % change vs. the nearest site along the gradient. **D**–**F** distance × intensity: % change per km for bats and moths at each intensity (points shaded from pale to deep purple for 10% to 100%). **G**–**I** distance × color: % change per km for bats and moths under red light (red points) and white light (white points with black outline). Each taxon's negative binomial GLMM (color × intensity + intensity × distance + color × distance + moon phase, non-forest cover, sky brightness, season, site random effect; sky brightness = the darkest of each site's lights-off sky photos, mean DSU); 95% CIs; rows share a y-axis. Asterisks mark estimates that differ from 0 (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Wald tests, Holm-adjusted within each taxon and panel in D–I).

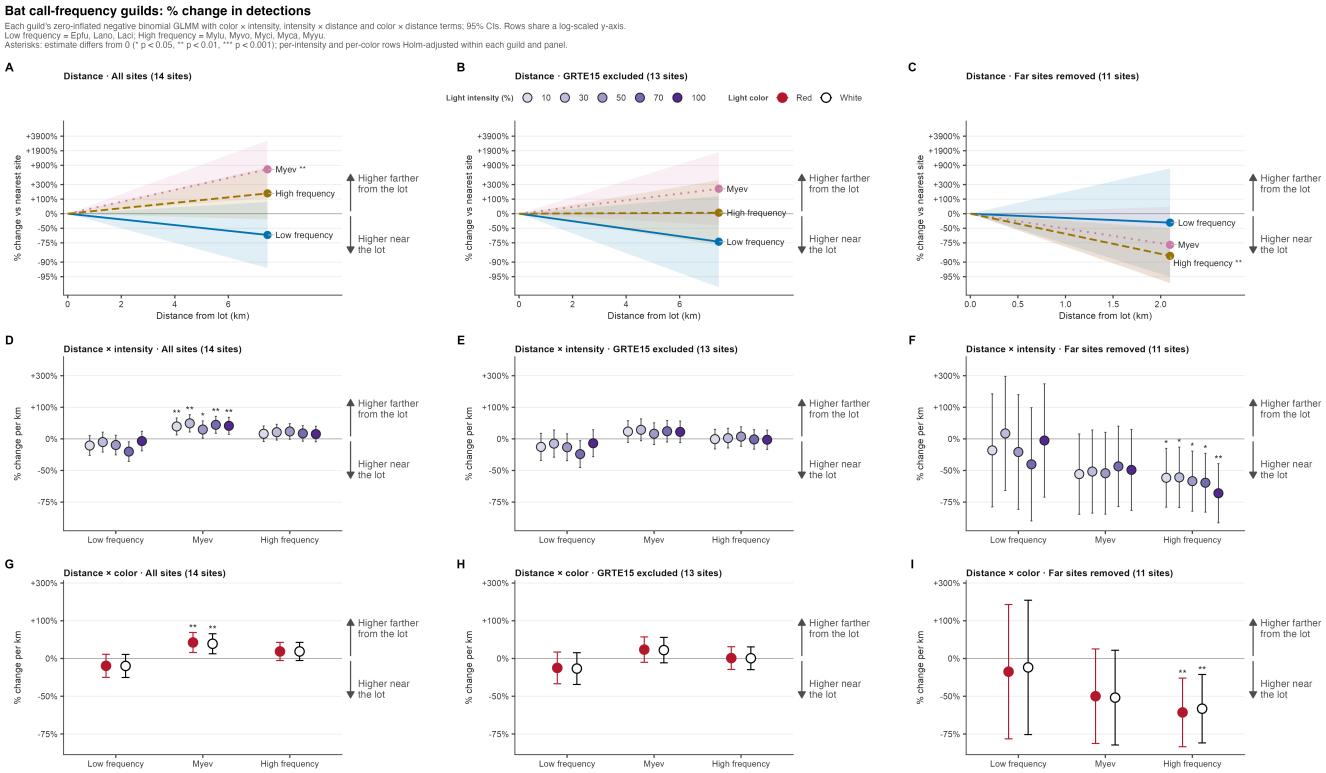


Figure 2: Bat call-frequency guilds: % change in detections. Low frequency (blue, solid) = big brown, silver-haired and hoary bats; Myev (pink, dotted) = long-eared myotis; high frequency (gold, dashed) = little brown, long-legged, small-footed, California and Yuma myotis. Columns, left to right: all 14 sites; all sites except GRTE15 (13 sites); far sites (GRTE13–15, ~7 km) removed (11 sites, 0–2.1 km). **A**–**C** distance: % change vs. the nearest site. **D**–**F** distance × intensity: % change per km for each guild at each intensity (points shaded from pale to deep purple for 10% to 100%). **G**–**I** distance × color: % change per km for each guild under red light (red points) and white light (white points with black outline). Each guild's zero-inflated negative binomial GLMM (color × intensity + intensity × distance + color × distance + moon phase, non-forest cover, sky brightness, season, site random effect); 95% CIs; log-scaled y-axis labeled in % change, shared within each row. Asterisks mark estimates that differ from 0 (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Wald tests, Holm-adjusted within each guild and panel in D–I).

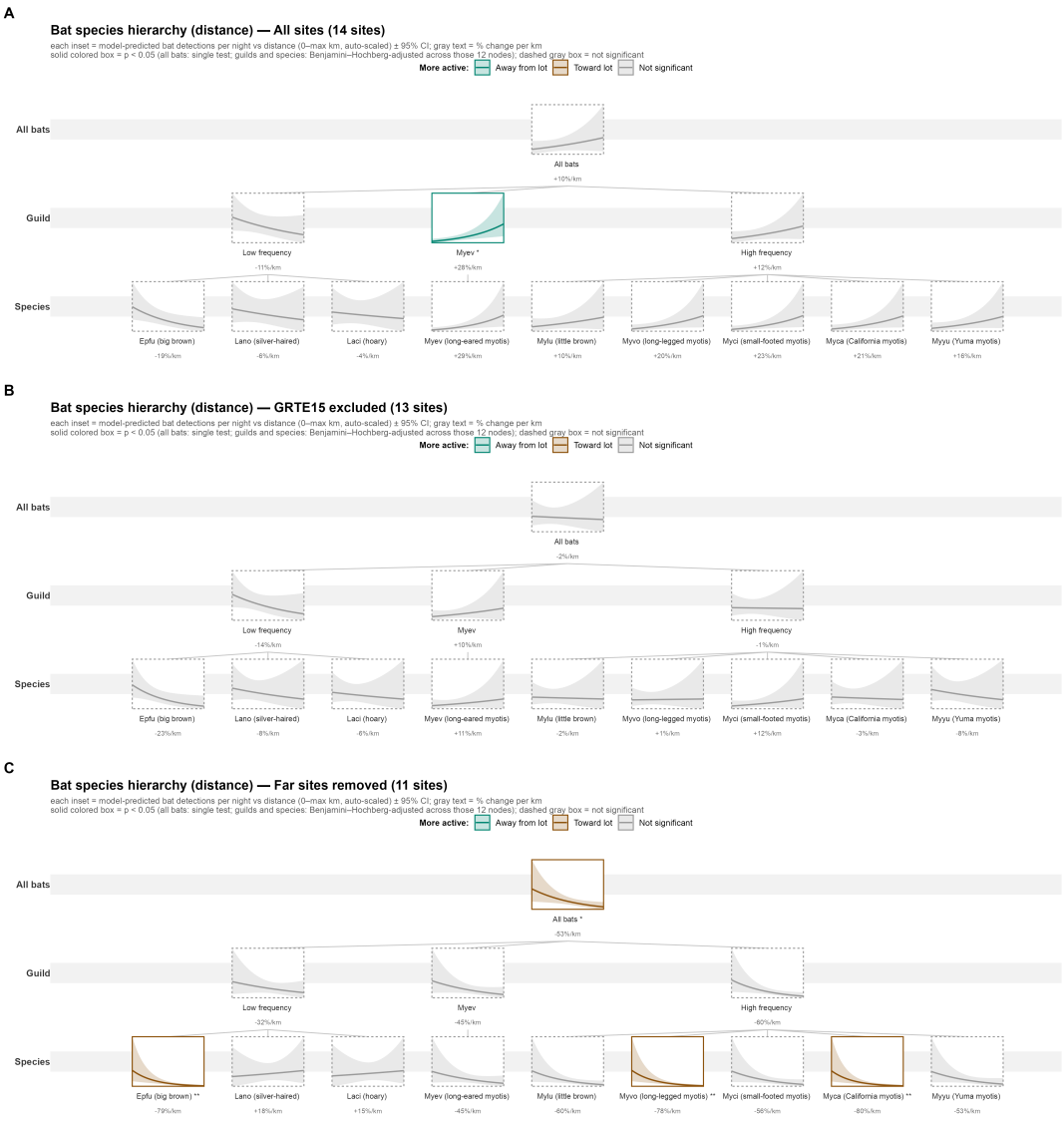


Figure 3: Bat distance responses arranged by tier (all bats → call-frequency guild → species), like the moth taxa figure. **A** all 14 sites; **B** all sites except GRTE15 (13 sites); **C** far sites removed (11 sites, 0–2.1 km). Each tier is a zero-inflated negative binomial GLMM of detections per site-night (taxon × distance + light color + intensity + moon phase, non-forest cover, sky brightness, season, site random effect; all bats = total detections, guilds and species = guild and single-species detections); the all-bats node is a single test (as in Figure 1) and is not adjusted; the 3 guild and 9 species nodes are Benjamini–Hochberg-adjusted together within each dataset. Unlike the moth figure, every node is drawn: a solid colored box marks a significant slope (teal = more active away from the lot, brown = more active near it; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$), a dashed gray box a non-significant one. Each inset is the model-predicted detections per night vs. distance (± 95% CI, auto-scaled); gray text = % change per km.

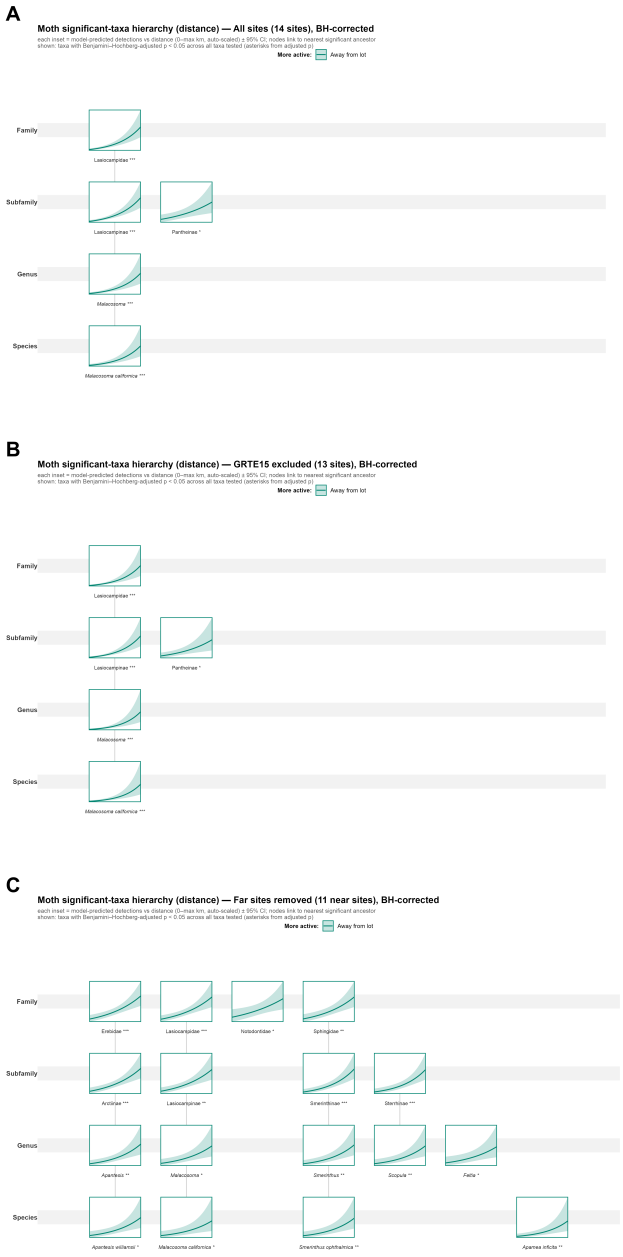


Figure 4: Moth taxa whose detections change significantly with distance from the lot after correcting for multiple testing, arranged by taxonomic level (family → subfamily → genus → species). **A** all 14 sites; **B** all sites except GRTE15 (13 sites); **C** far sites removed (11 sites, 0–2.1 km). Every taxon with ≥ 30 site-nights (51 taxa per dataset) was tested for a distance slope (taxon × distance negative binomial GLMM); shown are taxa with Benjamini–Hochberg-adjusted $p < 0.05$ across all 51 tests, each as a miniature of its model-predicted detections vs. distance (± 95% CI, auto-scaled within its box) and linked to its nearest significant ancestor. Asterisks from the adjusted p: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

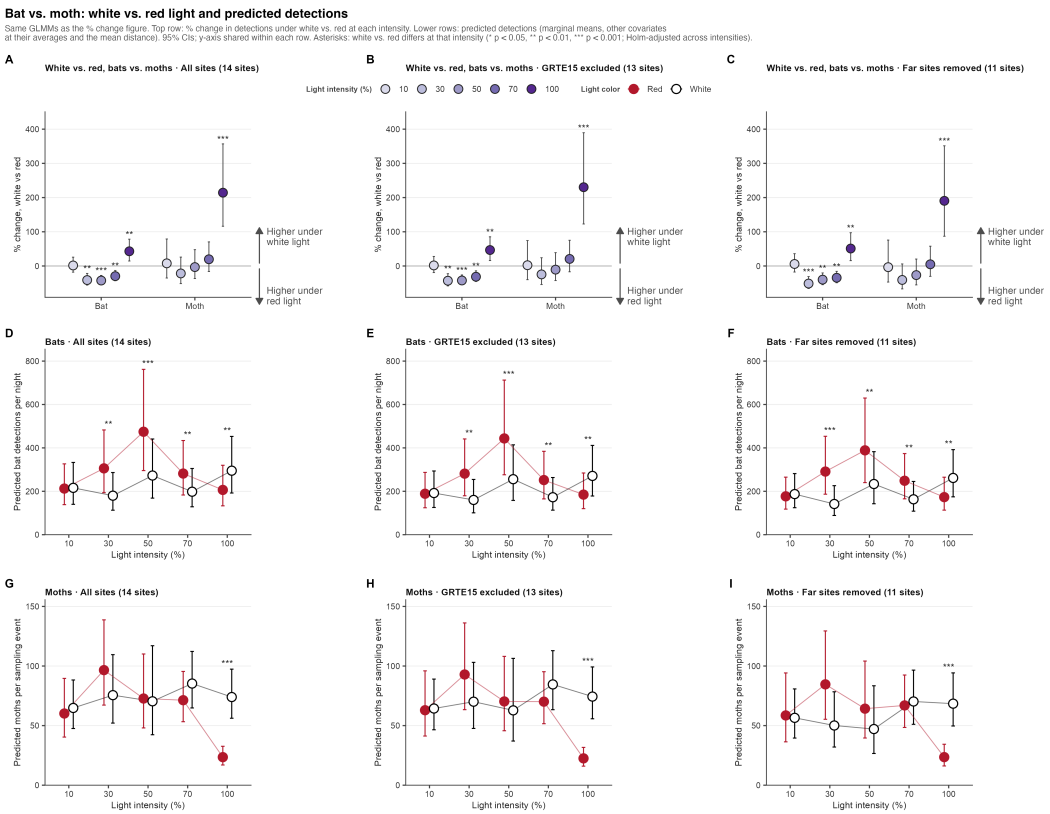


Figure 5: White vs. red light and predicted detections, from the same bat and moth GLMMs as the bat vs. moth % change figure (Figure 1). **A**–**C** % change in bat and moth detections under white vs. red light at each intensity (points shaded from pale to deep purple for 10% to 100%; contrasts at the mean distance). **D**–**F** predicted bat detections per night and **G**–**I** predicted moths per sampling event under red light (red points) and white light (white points with black outline) (marginal means with other covariates at their averages and the mean distance). Columns, left to right: all 14 sites; all sites except GRTE15 (13 sites); far sites removed (11 sites). 95% CIs; y-axis shared within each row. Asterisks mark intensities where white and red differ (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Holm-adjusted across the five intensities within each taxon).

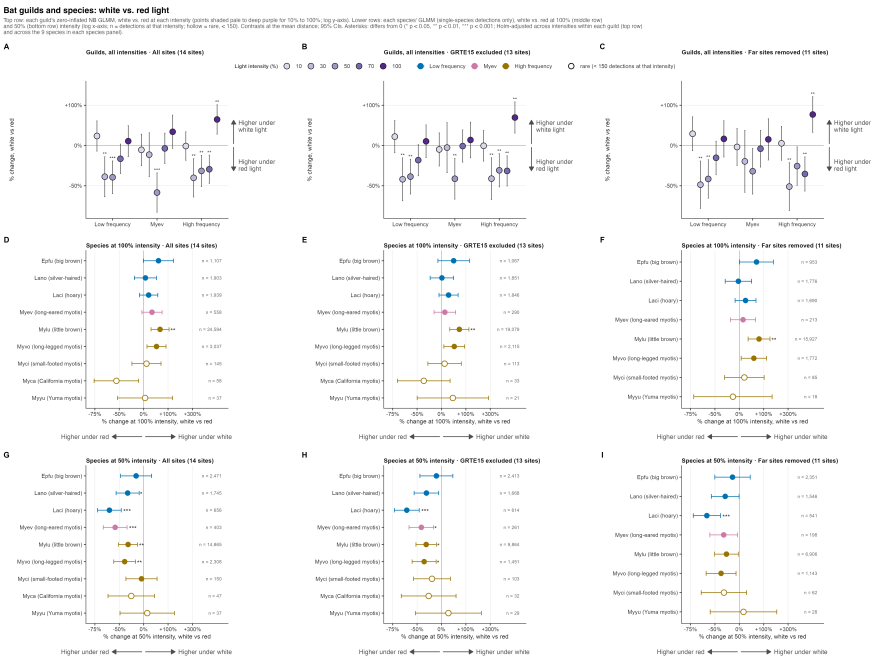


Figure 6: Bat call-frequency guilds and species: % change in detections under white vs. red light. **A**–**C** each guild (low frequency, Myev, high frequency on the x-axis) at each intensity (points shaded from pale to deep purple for 10% to 100%), from the guild zero-inflated negative binomial GLMMs above; log y-axis labeled in % change. **D**–**F** each species at 100% intensity and **G**–**I** at 50% intensity, from per-species GLMMs with the same structure (single-species detections only), colored by guild; log x-axis shared across D–I; n = detections at that intensity; hollow points are rare species (< 150 detections at that intensity). Columns, left to right: all 14 sites; all sites except GRTE15 (13 sites); far sites removed (11 sites). Contrasts at the mean distance; 95% CIs. Asterisks mark differences from 0 (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; Holm-adjusted across the five intensities within each guild in A–C, and across the 9 species in each panel in D–I).

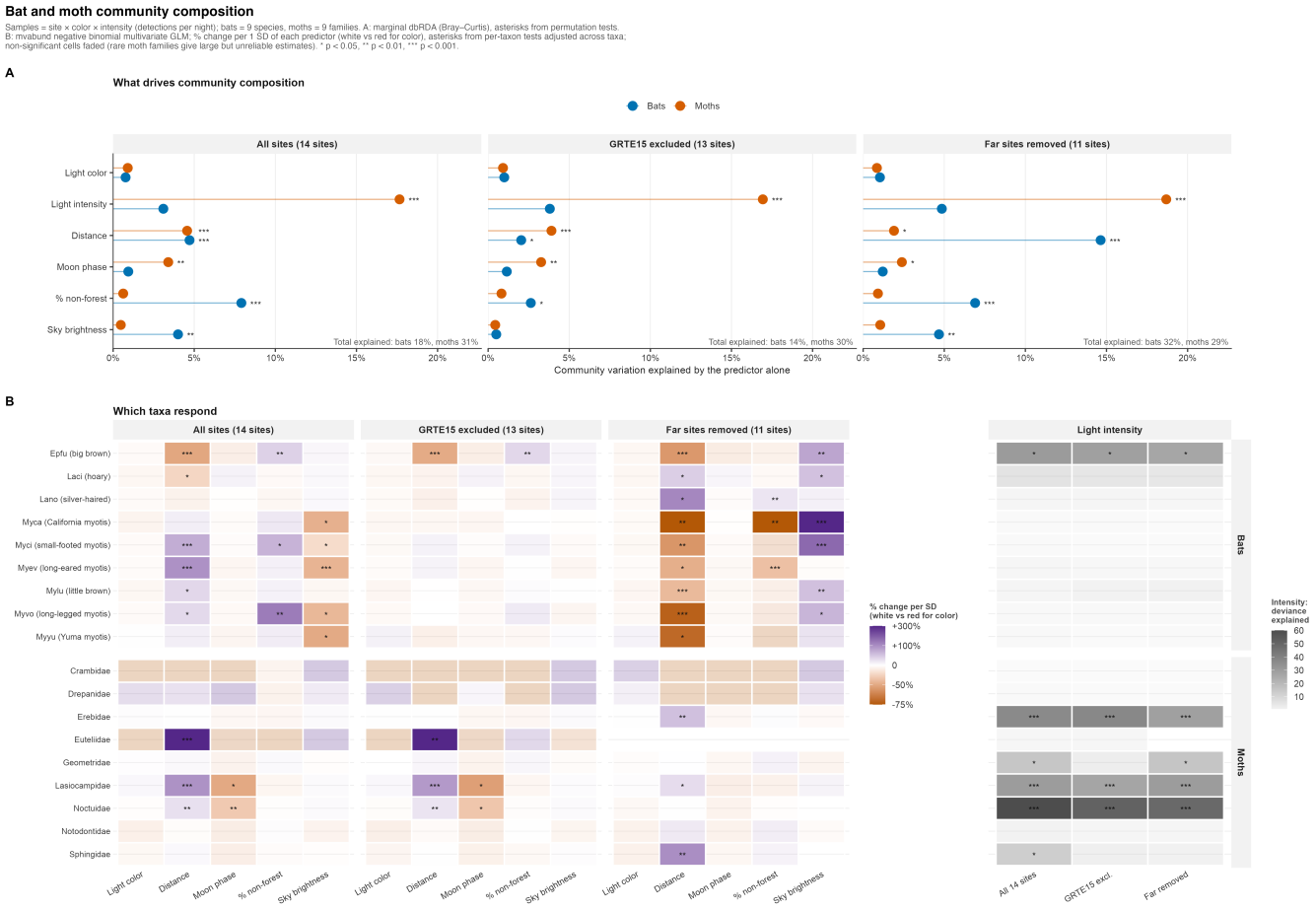


Figure 7: Bat (9 species) and moth (9 families) community composition, for all 14 sites, all sites except GRTE15 (13 sites), and with the far sites removed (11 sites). Samples are site × color × intensity combinations (detections per night). **A** Share of community variation explained by each predictor on its own (marginal dbRDA, Bray–Curtis distance), bats (blue) vs. moths (orange); asterisks from permutation tests. **B** Which taxa respond: % change in each taxon per 1 SD of each predictor (white vs. red for color), from a negative binomial multivariate GLM (mvabund), with intensity (five levels) shown separately as deviance explained (gray). Asterisks from per-taxon tests adjusted across taxa; non-significant cells are faded because rare moth families give large but unreliable estimates. Euteliidae was not caught at the 11 near sites. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.